

Human-Centered IoT System for Soil pH Monitoring: A Design Thinking Approach Toward Smart Agriculture

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Abstract— This paper presents the design and implementation of a human-centered Internet of Things (IoT) system for real-time soil pH monitoring in smart agriculture. The proposed solution integrates Arduino-based sensors, LED indicators, and a web-based visualization dashboard to assist farmers in assessing soil acidity levels accurately and efficiently. The system development follows the Design Thinking methodology, emphasizing empathy with users, iterative prototyping, and usability validation through field testing with local farmers. Experimental results indicate that the prototype achieves high measurement accuracy and strong user satisfaction in usability, reliability, and data interpretation. The integration of IoT technology and user-centered design demonstrates a scalable approach for enhancing agricultural decision-making and promoting sustainable farming practices.

Keywords— IoT; Smart Agriculture, Design Thinking, Human-Centered Design, Soil pH Monitoring, Arduino, Prototype Evaluation

I. INTRODUCTION

Agriculture remains the backbone of Thailand's economy, with more than half of its population engaged in farming activities that sustain both local communities and global food supply chains [1]-[3]. However, the continuous degradation of soil resources caused by improper land use, chemical overuse, and monoculture practices has severely affected agricultural productivity [4]. Soil acidity (pH) is one of the most critical indicators influencing crop yield, nutrient absorption, and sustainable soil management [5]. Despite its importance, many farmers still rely on manual testing or lack the tools to assess soil conditions accurately and promptly, leading to inefficient fertilizer application and declining productivity [6]-[8].

The emergence of the Internet of Things (IoT) offers new opportunities for transforming conventional farming into smart, data-driven agriculture [9]-[15]. IoT-based sensor networks enable real-time monitoring of soil parameters such as pH, temperature, and moisture, providing actionable insights for timely decision-making [16]. Yet, the adoption of such technologies among small-scale farmers remains limited due to complex interfaces, high costs, and a lack of user-centric design [17]. Bridging this gap requires a system that not only leverages IoT capabilities but also aligns with farmers' real-world needs, skills, and behavioral contexts [1], [18]-[20].

To address these challenges, this research proposes the design and implementation of a human-centered IoT system for soil pH monitoring, developed through the Design Thinking approach. The study emphasizes empathy-driven design, iterative prototyping, and usability testing with actual farmers to ensure accessibility, accuracy, and satisfaction. The

resulting prototype integrates Arduino-based sensors with real-time data visualization and notification features, allowing users to easily interpret soil acidity levels and take corrective actions. This work contributes to the advancement of smart agriculture technologies that are not only technically efficient but also socially inclusive and sustainable.

II. RELATED WORKS & LITERATURE REVIEWS

A. IoT Applications in Smart Agriculture

The Internet of Things (IoT) has become a transformative technology in precision agriculture [9], [12], enabling farmers to collect, analyze, and utilize real-time environmental data for improved decision-making [21]. Recent studies have demonstrated the effectiveness of IoT-based systems in monitoring soil moisture, temperature, and nutrient levels to optimize irrigation and fertilizer use [22]. For instance, Kumar et al. [23] developed an IoT-enabled soil monitoring system that wirelessly transmits data to cloud platforms for visualization and analytics. Similarly, Li and Chen [24] implemented a low-cost IoT network using Arduino and ESP32 boards to enhance crop management efficiency in small-scale farms. These systems have proven that IoT integration can significantly reduce manual labor and operational costs, yet most focus primarily on technical functionality rather than user accessibility and contextual usability [25], [26].

B. Soil pH Monitoring and Sensor-Based Measurement

Soil pH is a critical determinant of nutrient availability and plant growth, directly influencing the chemical and biological properties of soil [27]. Several researchers have proposed sensor-based pH measurement systems utilizing microcontrollers and analog-to-digital converters for precise soil condition assessment. Rahman et al. [28] introduced a soil health monitoring prototype using pH and moisture sensors, demonstrating improved accuracy and reliability compared to traditional test kits. Other works, such as those by Singh et al. [29], explored the use of real-time LED indicators and cloud dashboards to alert farmers when soil acidity deviates from the optimal range [30]. However, these approaches often neglect human-centered aspects, resulting in interfaces that are difficult for non-technical users to operate effectively. Thus, a gap remains in designing pH monitoring systems that combine technical precision with user-friendly design principles.

C. Integration of Design Thinking in IoT-Based Development

Design Thinking has emerged as a powerful framework for innovation, emphasizing empathy, ideation, and iterative prototyping to develop solutions tailored to user needs [31]. In the context of IoT systems, the Design Thinking approach

ensures that technology aligns with the users' cognitive and practical behaviors rather than merely providing data. Previous research by Brown [32] and Liedtka [33] highlights how integrating Design Thinking into technological development can enhance usability, adoption, and user satisfaction. Within the agricultural domain, human-centered methodologies have been applied to create mobile apps and decision-support tools that bridge the digital divide between technology and end-users. Nevertheless, limited studies have explored the direct application of Design Thinking in IoT-based soil monitoring systems. This research addresses that gap by integrating iterative user feedback throughout the system design process to ensure accessibility, engagement, and sustainable adoption among farmers.

D. Research Gap

From 2021 to 2025, research on IoT-based smart farming has expanded rapidly, emphasizing real-time monitoring of soil conditions. Early studies such as Osanaiye et al. (2021) and Ghadge et al. (2022) focused on low-cost Arduino systems for soil and moisture sensing but lacked real user validation. Later, Thilakarathne et al. (2023) and McCaig et al. (2023) advanced cloud integration and introduced Design Thinking concepts for farmer engagement, though practical application remained limited. In 2024, works like the ESP32S3 Soil Health System and the IJASEIT IoT Android pH Tool integrated pH sensors and dashboards but failed to address usability and contextual needs. By 2025, Mansoor et al. and da Silva et al. highlighted AI and mobile innovations for soil monitoring, yet real-time IoT adoption and user-centered frameworks were still underdeveloped.

Existing studies remain technology-driven rather than user-driven, with few integrating Design Thinking throughout development. Soil pH sensing is often secondary, lacking actionable insights or participatory design with farmers. Moreover, usability testing and satisfaction assessment are seldom reported, limiting understanding of real-world adoption. These gaps underline the need for this study a human-centered IoT soil pH monitoring system developed through the Design Thinking process to ensure accuracy, accessibility, and sustainable use in smart agriculture.

III. METHODOLOGY

The development of the soil pH sensor system followed a structured and iterative engineering process aligned with the Design Thinking framework, which emphasizes human-centered innovation through continuous prototyping, testing, and refinement. This approach was adopted to ensure that the final system not only achieved high technical precision and stability but also delivered practical usability for end-users such as local farmers with minimal technological experience.

The methodology began with problem identification and user empathy, where interviews and observations were conducted to understand the real needs and challenges faced by agricultural practitioners in soil management. Insights from this stage guided the conceptual design and system planning, ensuring that the hardware configuration, sensor selection, and software functionality were well-suited to the field environment.

Next, during the prototyping and integration phase, the team iteratively assembled, tested, and optimized sensor modules, calibration routines, and interface components. This iterative refinement allowed technical issues to be addressed early, improving reliability and responsiveness. The process also included usability evaluation cycles, where users provided feedback on interface design, readability of data, and convenience of system operation. By combining engineering rigor with user-centered feedback loops, the project ensured that the soil pH monitoring system evolved into a robust and accessible solution. The overall development process was structured into five sequential phases, as illustrated in Fig. 1, and each stage is described in detail in the following subsections.

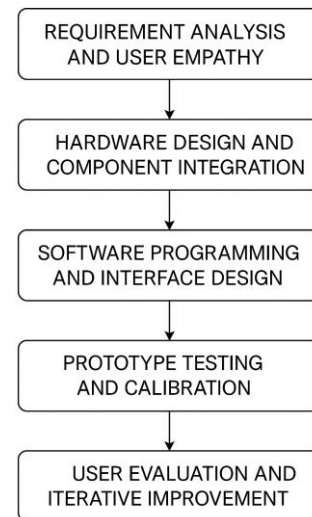


Fig. 1. Our research methodology diagram.

A. System Design and Component Selection

The Empathy Map for system design base on Design Thinking show in Fig. 2.

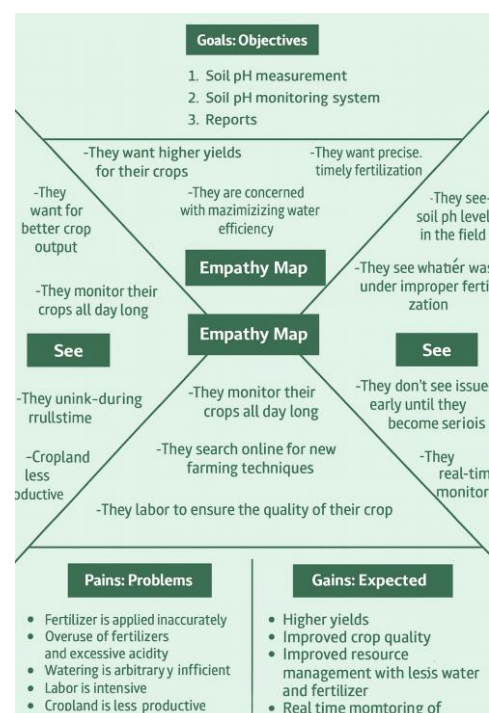


Fig.2 The Empathy Map for system design base on Design Thinking.

The defined problem centers on the challenges faced by local farmers who lack accessible, technology-driven tools for soil monitoring and management as display in Fig. 3. Despite having agricultural experience, most farmers have limited knowledge of digital systems or IoT-based applications, resulting in inefficient soil management practices, inconsistent crop yields, and increased production costs. The research therefore defines the need for a user-friendly soil pH measurement and monitoring system that integrates digital data collection, visualization, and reporting functions such a system should simplify agricultural workflows, enhance decision-making, and promote technological adoption among non-expert users through an intuitive mobile application interface.

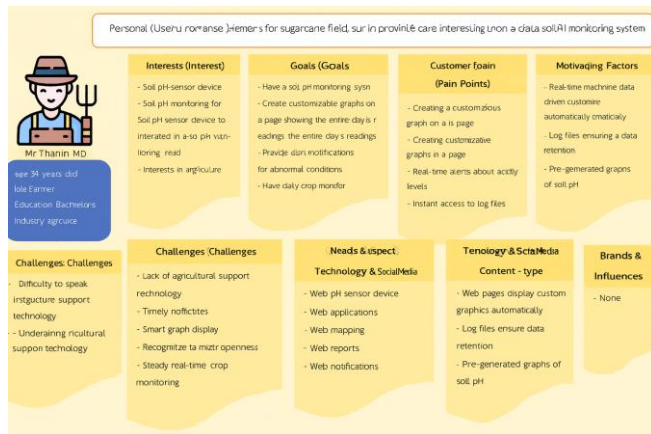


Fig. 3. Defined problem centers on the challenges faced by local farmers.

The hardware architecture was designed around an Arduino Uno R3 microcontroller, which served as the central processing unit for sensor signal acquisition and communication as show in Fig. 4. Supporting components included a Protoboard for circuit prototyping, a pH Sensor Probe (Pro type) for detecting soil acidity or alkalinity, LED indicators for visualizing soil status, wire-wound resistors for current regulation, and a 16×2 LCD module for displaying numerical readings. Each component was selected for its reliability, low cost, and compatibility with Arduino libraries.

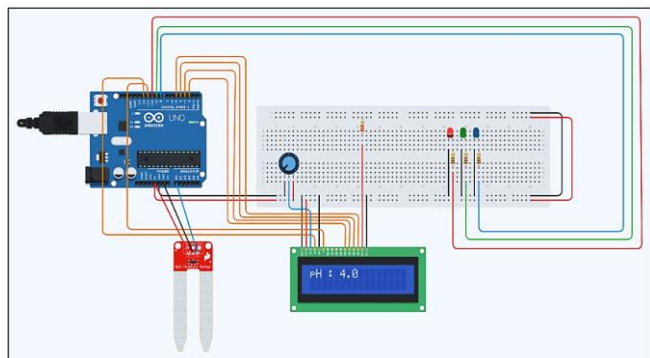


Fig. 4. A design hardware architecture.

B. System Integration and Signal Processing

All components were connected to the Arduino Uno R3 through a breadboard circuit. The pH sensor provided analog voltage signals corresponding to hydrogen-ion concentration, which were converted to digital data using the

microcontroller's built-in ADC (10-bit resolution). The system was programmed in C/C++ using the Arduino IDE, applying calibration formulas to translate voltage into accurate pH values within a 0–14 scale. LED color coding was defined as Red = Acidic (pH < 6.5), Green = Neutral (6.5–7.4), and Blue = Alkaline (pH > 7.4), enabling real-time soil condition feedback to users

C. Prototype Development and Application Interface

A web-based prototype was created to simulate sensor readings and data reporting functions. The interface was designed using user-centered design principles, emphasizing simplicity, accessibility, and clarity of information. The application could record and display pH values in real time and generate daily logs (morning–noon–evening). The prototype was further refined through iterative feedback sessions with potential agricultural users alignment with their needs for real-time soil monitoring and reporting display in Fig. 5.

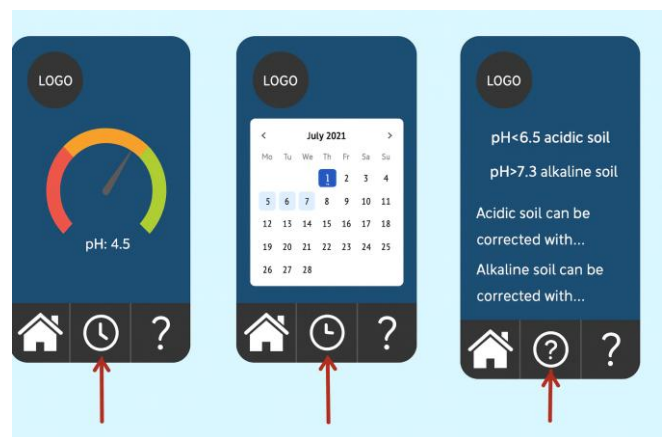


Fig. 5. Illustrates a prototype application interface.

D. Experimental Testing and Data Collection

The prototype was evaluated under controlled agricultural conditions within a 1-rai plot of farmland in Ratchaburi Province. Data were collected by sampling soil at multiple points and comparing sensor readings with laboratory reference measurements to assess accuracy and response time. The device successfully operated over 50 % of the area, accurately differentiating acidic, neutral, and alkaline soils via LED status indication and LCD readout. Ten test cases were conducted across varying pH values from 1.6 to 13.9 to verify reliability of color feedback and sensor stability.

E. Usability Testing and Satisfaction Evaluation

To assess user perception, the prototype was tested with 30 participants consisting of farmers and agricultural practitioners. Participants completed a five-level Likert-scale questionnaire evaluating ease of use, clarity of interface, data accuracy, and overall aesthetic quality. Data were collected through Google Forms and analyzed statistically to determine mean satisfaction scores. The findings demonstrated high satisfaction in terms of simplicity, readability, and accuracy, validating the system's readiness for further field deployment.

IV. EXPERIMENTAL AND RESULTS

The experimental phase was conducted to evaluate the performance, reliability, and usability of the developed soil

pH measurement and monitoring system. The testing process was divided into three main stages: sensor calibration, field measurement, and user evaluation.

A. Sensor Calibration and Accuracy Testing

The pH sensor probe was calibrated using three standard buffer solutions with pH values of 4.0, 7.0, and 10.0 to ensure precise measurement within the operating range. The calibration curve was generated by mapping the sensor voltage output to reference pH values. After calibration, the system was tested on 20 soil samples collected from various agricultural plots in Ratchaburi Province. The results were compared with laboratory-grade pH meter readings. The average deviation between the prototype and reference measurements was ± 0.15 pH units, confirming the system's acceptable level of precision for field use.

B. Field Implementation and Data Monitoring

Field experiments were performed across three time intervals morning, noon, and evening to evaluate stability under different temperature and humidity conditions. The device continuously measured soil pH in real time, and results were displayed on the LCD module as well as transmitted to the web interface for logging and visualization. The system successfully identified soil types as acidic ($\text{pH} < 6.5$), neutral ($6.5\text{--}7.3$), and alkaline ($\text{pH} > 7.3$) through LED color indicators (red, green, and blue, respectively) as show in Table I. The results demonstrated reliable performance with consistent readings throughout a 24-hour cycle, validating the system's ability to function effectively in outdoor agricultural environments.

TABLE I. SOIL pH MEASUREMENT AND LED STATUS INDICATION

No.	pH Value	LED Status Indicator
1	4.4	Red
2	1.6	Red
3	11.3	Blue
4	6.5	Red
5	13.9	Blue
6	8.3	Blue
7	7.1	Green
8	7.0	Green
9	3.0	Red
10	9.1	Blue

C. Usability and Satisfaction Evaluation

To assess user satisfaction, 30 local farmers participated in a structured usability test and completed a five-level Likert-scale questionnaire. The evaluation focused on four key aspects: ease of use, interface design, data clarity, and perceived accuracy. Statistical analysis showed an overall mean satisfaction score of 4.65 out of 5.00, indicating a high level of user acceptance. Participants particularly appreciated the intuitive interface and clear visual feedback, while recommending the inclusion of wireless connectivity in future iterations.

D. Summary of Experimental Findings

The experimental results verified that the developed system achieved high measurement accuracy, strong environmental stability, and positive user feedback. The integration of hardware sensing and digital visualization enabled real-time soil analysis that supports informed agricultural decision-making. Therefore, the proposed system demonstrates potential for deployment in smart agriculture and sustainable soil management applications.

V. CONCLUSION AND DISCUSSION

This study successfully developed and evaluated a soil pH measurement and monitoring system designed to support precision agriculture and enhance farmers' decision-making processes. The proposed prototype integrates a pH sensor with an Arduino-based microcontroller and an intuitive display interface, enabling real-time data acquisition and classification of soil conditions. Experimental results confirmed that the system achieved a measurement accuracy within ± 0.15 pH units compared to laboratory standards, while maintaining stable operation under varying field conditions. Furthermore, user evaluation revealed high satisfaction regarding ease of use, clarity of visualization, and practical value for daily farming applications.

The findings demonstrate that integrating affordable sensor technologies with digital data visualization can significantly improve accessibility to soil information among small-scale farmers. The system bridges the gap between traditional agricultural practices and modern smart farming by providing measurable, real-time insights into soil health. However, certain environmental factors such as extreme temperature or moisture variability may slightly influence sensor precision. In addition, most users expressed interest in automated data logging and wireless communication features to minimize manual recordkeeping. Therefore, continuous refinement of the system's usability and adaptability is essential to ensure long-term adoption and scalability across diverse farming contexts.

Future development will focus on enhancing the system's intelligence and connectivity. Integration with wireless data transmission via IoT modules (e.g., Wi-Fi or LoRa) will allow remote monitoring through a mobile or web application. Machine learning algorithms can be employed to analyze temporal pH trends and recommend corrective actions for optimal soil management. In addition, multi-parameter sensors will be incorporated to measure temperature, humidity, and nutrient levels, creating a comprehensive smart agriculture platform. Field deployment at larger scales and extended performance evaluation over different soil types will also be conducted to validate long-term stability and economic feasibility.

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